
Date: 16 August 2018

SECTION 1 GENERAL

1.1 OBJECTIVE

This Technical Circular (TC) provides guidance to Airport Operators on the preparation and contents of an Aerodrome Manual (AM) for aerodrome operations in line with SARPS in Annex 14 Part I, MOZCAR Parts 139 and MOZCATS 139, Vol. I.

1.2 APPLICABILITY

This CT apply to all Aerodrome Operators, who operate in Mozambique according prescribed on MOZCAR 139, Subpart VI.

1.3 REFERENCES

- 1) MOZCAR Part 139
- 2) MOZCATS Part 139, Vol. I and II
- 3) Manual of Aerodrome Certification Procedure
- 4) CIA - Processo das Cinco Fases de Certificação de Aeródromo
- 5) ICAO Doc 9774 - Manual on certification of aerodromes
- 6) CT 100-001 Statement of Compliance with Regulatory Requirements

1.4 CHANGES

- 1) This is an original issuance of this CT.
- 2) This CT cancels Document Number: IACM-AGA-301-04 – Aerodrome Manual Template, First Edition

1.5 BACKGROUND

1.5.1 ICAO Standards in Annex 14, MOZCAR 139 and MOZCATS 139 require an Aerodrome Operator to establish an Aerodrome Manual, in accordance with the specified requirements, containing all pertinent information on the aerodrome site, facilities, services, equipment, operating procedures, organization and management including a safety management system to guide operational staff in the performance of their duties.

1.5.2 In order to obtain IACM Certificate/Licence, the Aerodrome Operator must ensure that the manual follows the organisation, structure and include, as a minimum, the contents specified in the governing regulation, as applicable to the operation.

1.5.3 Aerodrome Operators are required to amend or revise the AM, as necessary, to ensure that the information contained therein is kept up-to-date, by reflecting any changes to the regulatory and IACM requirements or to the operation. All such amendments or revisions must be submitted to the IACM for approval prior to implementation and issued to all personnel that are required to use the AM.

SECTION 2 – AM APPROVAL PROCESS

2.1. Initial approval of the AM

2.1.1. During the initial Aerodrome Operator's Certification/Licensing process operators are required to submit the AM to the IACM for approval.

2.1.2. The application should contain:

- i) the Checklist "Form number 13 - Aerodrome Manual Compliance Checklist";
- ii) two originals of the AM (may be presented in one or more volumes);
- iii) a Statement of Compliance with the applicable regulations MOZCAR Part 139.

2.1.3. Upon receipt, the IACM performs a cursory review of the application and informs the applicant, in writing, of its acceptability. Incomplete applications are returned with an explanation of the reason for rejection.

2.1.4. A detailed review of the AM is performed by comparison of its contents with the applicable regulatory requirements, considering all aspects of the proposed operations. The Aerodrome Operator is notified of any deficiencies for correction;

2.1.5. Upon being satisfied that all deficiencies identified during the manual review process and during the demonstration and inspection phase have been corrected and ascertaining that the manual contents properly addresses all applicable regulatory requirements and correctly reflects the operator's organization and operation, in conformity with the approvals and authorisations sought in the operator's operations specifications, the IACM issues the approval of the AM.

2.1.6. The IACM approval of the AM is reflected by the stamping and signature on the Airport Operator Statement and Aerodrome Manual Approval and list of effective pages of the two original manuals.

2.1.7. The IACM keeps one original of the approved AM and returns the other original to the applicant.

2.2. Amendment and revision of the AM

2.2.1. The Aerodrome Operator is required by regulation to keep the information in the AM up-to-date. The AM should be amended whenever a change occurs in the requirements or in the operation.

2.2.2. All amendments and revisions to the AM, except urgent temporary revisions required in the interest of safety, where a submission must be made with 7 days of adoption, must be approved by the IACM prior to implementation.

2.2.3. The application for an amendment or revision should contain:

- i) cover letter explaining the proposed changes;
- ii) two originals of the proposed amendment or revision, including the amended pages with highlights of the proposed changes, as per the revision procedures, and updated List of Effective Pages and Table of Contents;
- iii) if affected, an updated version of the existing Statement of Compliance with the applicable regulations MOZCAR 139.

2.2.4. The approval process follows, with the necessary, adaptations the steps described above for the initial approval.

SECTION 3 – ORGANISATION, STRUCTURE AND CONTENTS OF THE AM

3.1. Organisation Of The AM

3.1.1. When preparing an AM the operator should ensure that the policies and procedures contained in the manual attain the following objectives:

- i) implement, and are not contrary to, the Civil Aviation Regulations of Mozambique (MOZCAR);
- ii) do not contravene the terms and conditions in the Aerodrome Operator Certificate and associated operations specifications;
- iii) provide clear, complete and detailed operating instructions, policies and procedures so that operational personnel are fully informed of what is required of them. Procedures shall be effective, represent sound safety philosophy and be capable of being accomplished;
- iv) make provisions for revision to ensure that the information contained therein is kept up-to-date; and
- v) present the necessary guidance and instructions to personnel in a suitable and convenient easy to read, easy to amend format.

3.1.2. The AM may be complemented by other manuals, covering specific subjects. The size, as well as the number of volumes of the AM will depend upon the size and complexity of the proposed activities.

3.1.3. In all cases, the following areas must be considered:

- i) Organization and readability. The manual(s) shall be organized so that information specific to various employee positions and types of operations is easy to locate, clear, concise, and unambiguous. Tables of contents shall be detailed enough so that specific subject areas may be easily and expeditiously located. Print quality, illustrations, and graphics shall be clear and readable. Each manual and appendix shall be numbered and issued according to a specific distribution list, and each holder made responsible for its prompt and accurate update. The distribution list shall contain all operations personnel and others requiring the information therein for proper performance of their duties.
- ii) Validity and accuracy. Technical information contained in manuals such as coordinates data, laboratory data, suppliers' certificates, etc. shall accurately reflect data provided from the source or shall have been developed through the use of accepted and approved methods;

- iii) Consistency. Information presented in the various sections or volumes of the manual shall be consistent with that presented in other sections;
- iv) Currency and conformity. Information contained in the manual shall reflect current company organization, equipment, procedures and policies. The manual(s) shall be easy to update and contain a list of effective pages;
- v) Distribution and availability. The operator shall have an effective system for distributing and updating manuals. The individual(s) responsible for entering changes in specific manuals shall be identified. The IACM must be provided with copies of all manuals;
- vi) Content. The AM Compliance checklist form number 13 (in attach) shall be used by the operator to ensure that all subject areas are adequately addressed in the operator's manual. Certain items may not apply to a particular operator in which case the checklist item shall be annotated – not applicable. More specific information on each checklist item is outlined below. This same checklist is used by IACM inspectors to determine the acceptability of the material contained in the manual(s).

3.1.4. The AM shall contain the Aerodrome Operator's organization procedures. It states the organization management team and the organization commitment to comply with the regulatory requirement and to maintain the standards established during the approval certification process.

3.1.5. It explains in details the operator's responsibilities, functions and obligations. It further explains the regulatory process, methods, procedures and capabilities the operator employs to satisfy these regulatory requirements.

3.1.6. The AM defines the aerodrome operator's management structure, quality system management, maintenance activity coordination, duties and responsibilities of management personnel.

3.2. Structure of the AM

The AM shall contain, as a minimum, the following sections, including some of their requirements:

- I. a Airport Operator Statement and Aerodrome Manual Approval;
- II. a Distribution List;
- III. a Record of Amendments;

- IV. a Exemptions Record;
- V. a List of Pages in Effect;
- VI. a Table of Contents;

CHAPTER 1 – GENERAL

This part shall include:

- 1.1 The Object and Scope of The Aerodrome Manual;
- 1.2 Law and Regulations Requirements;
- 1.3 Airport Operating Conditions;
- 1.4 Aeronautical Information System and Publishing Procedures;
- 1.5 Aircraft Movements Recording System;
- 1.6 Airport Operator's Obligation;
- 1.7 Use of Common Reference Systems; and
- 1.8 Definitions and Acronyms.

CHAPTER 2 - INFORMATION ON AIRPORT LOCATION

This part shall comprise all Airport Plans.

CHAPTER 3 - AIRPORT INSTALLATIONS

This part shall include:

- 3.1. Airport Data;
- 3.2. Airport Dimension Data.

CHAPTER 4 - OPERATIONAL PROCEDURES AND AIRPORT SAFETY MEASURES

This part shall include:

- 4.1. Airport Reporting;
- 4.2. Access to the Airport Movement Area;
- 4.3. Airport Emergency Plan;
- 4.4. Rescue and Fire-Fighting Services;
- 4.5. Inspection of the Movement Area;
- 4.6. Maintenance of The Movement Area;
- 4.7. Visual Aids and Airport Electrical System;
- 4.8. Airport Works Safety;
- 4.9. Apron Management;

- 4.10. Apron Safety Management;
- 4.11. Airside Vehicle Control;
- 4.12. Wildlife Hazard Management;
- 4.13. Obstacle Control;
- 4.14. Removal of Disabled Aircraft;
- 4.15. Handling of Hazardous Materials;
- 4.16. Low Visibility Operations;
- 4.17. Protection of Sites for Radar, Navigational Aids and Meteorological Equipment

CHAPTER 5 - AIRPORT ADMINISTRATION

This part shall comprise the Information on Airport Administration.

CHAPTER 6 - SAFETY MANAGEMENT SYSTEM (SMS)

This section shall include:

- 6.1. Component 1. Safety Policy and Objectives;
 - i) Element 1. Management commitment and responsibility;
 - ii) Element 2. Safety Accountabilities;
 - iii) Element 3. Appointment of Key Safety Personnel;
 - iv) Element 4. Coordination of Emergency Response Planning; and
 - v) Element 5. SMS Documentation.
- 6.2. Component 2. Safety Risk Management
 - i) Element 1. Hazard Identification; and
 - ii) Element 2. Safety Risk Assessment and Mitigation
- 6.3. Component 3. Safety Assurance
 - i) Element 1. Safety Performance Monitoring and Measurement;
 - ii) Element 2. The Management of Change; and
 - iii) Element 3. Continuous Improvement of the SMS
- 6.4. Component 4. Safety Promotion
 - i) Element 1. Training and Education;
 - ii) Element 2. Safety Communication

CHAPTER 7 - AIRPORT OPERATIONS PROTOCOLS, ACCORDS, CONTRACTS AND CONVENTIONS WITH THIRD PARTIES

This part shall comprise the Airport operations protocols, accords, contracts and conventions with third parties shall be included in the Aerodrome Manual.

CHAPTER 8 - APPENDICES

This part shall include:

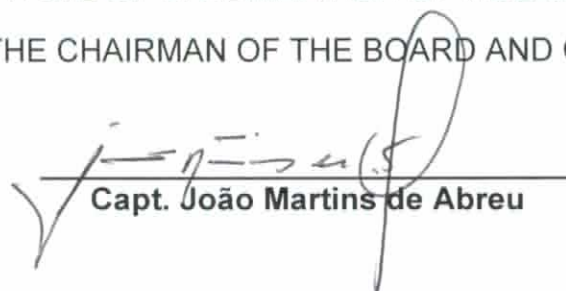
- 6.1) Plans
- 6.2) Airport Emergency Plan
- 6.3) Other Appendices.

3.3. The Contest of the AM

3.3.1. The contents of the AM shall cover all Parts, Titles, Paragraphs and subparagraphs as defined in MOZCAR 139, Subpart VI and MOZCATS 139, Vol. I.

3.3.2. The Operators shall use the Compliance checklist form number 13 (in attach) to ensure that all pertinent information is included in each Part of the AM. Additionally, find a Template of Aerodrome Manual in appendix 1.

INSTITUTE OF CIVIL AVIATION OF MOZAMBIQUE
THE CHAIRMAN OF THE BOARD AND CEO



Capt. João Martins de Abreu

NAME OF AERODROME OPERATOR

AERODROME OPERATOR LOGO

AERODROME MANUAL COMPLIANCE CHECKLIST

Note 1: The Team Leader may delegate the assessment of development of this aerodrome manual in full or in parts to any other member of team. Nevertheless the Team Leader remains responsible. On completion of the assessment, the filled check list shall be enclosed to this Aerodrome Manual and sent together to IACM for the purpose of assessment and to be included in the aerodrome certification file.

Note 2: The details listed represent the minimum content of an aerodrome manual. The Team Leader must bear in mind that the content of this aerodrome manual must reflect the complexity and operating environment of this particular aerodrome.

Name of Aerodrome: _____

Checked by: _____

Name of Team Leader: _____

Revised by: _____

Date: _____

Date of Approval: _____

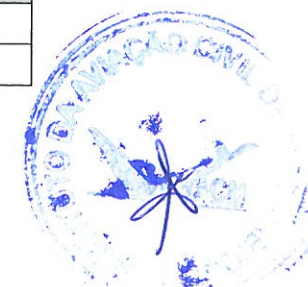
AMENDMENT OF AERODROME MANUAL NUMBER _____



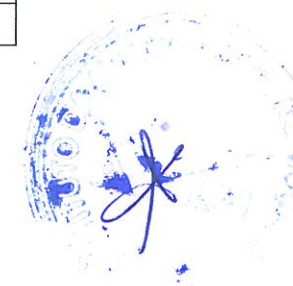
Item	Requirement	Yes	No	Reference Pg. and Comments
1. Writing, presentation and administration				
1.	Is the Aerodrome Manual (AM) signed by the Aerodrome Operator's Representative?			
2.	The first pages of the AM: • AO's Statement and AM Approval? • AM Distribution list? • Record of amendment? • List of effective pages? • Table of contents? • List of Appendices?			
3.	Page heading centered, showing the title of the manual and the name of the aerodrome?			
4.	Page footing showing:: • The edition & revision number and the date it came into effect? • The current page number and the total number of pages?			
5.	Reference publications or documents not included, but being part of the AM published separately?			
6.	Is the format of the AM A4 paper size?			
7.	AM easiness of the insertion or the removal of pages when updating?			
8.	AM eight (8) chapters identified with dividers?			
9.	Description of a procedure for regular amendment whenever necessary to maintain its currency			
10.	Procedure to ensure that IACM is advised in writing within 14 days of any amendment of the AM			
11.	Can a reader tell if the AM is up to date?			



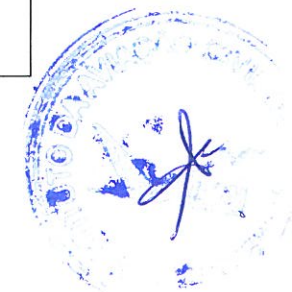
Item	Requirement	Yes	No	Reference Pg. and Comments
Part 1: General				
12.	AM specifies the object and scope?			
13.	AM specifies Aerodrome Certification laws and regulations?			
14.	Conditions applicable to the use of the aerodrome.			
15.	Description of the aeronautical information process and the publishing procedures in force at the aerodrome.			
16.	Description of the aircraft movement recording system applied at the aerodrome.			
17.	Description of the Aerodrome operator's obligations.			
18.	Definitions			
19.	Acronyms			
Part 2: Information on aerodrome location				
20.	Details of the land certificate(s) for the aerodrome site or details of the control over (eg, lease agreements) the property on which the aerodrome is located?			
21.	Plan showing the main operational installations, such as the wind direction indicators, the PAPIs, etc.?			
22.	Plans showing the delimitation between the aprons and the maneuvering area, the cargo area, the terminal building, the navigational aids and the meteorological installations?			
23.	Plan (to scale) showing the apron with aircraft stands marking?			
24.	Plan (to scale) showing the maneuvering area?			
25.	Plan showing the details of runway lighting, the PAPIs, markers and signs?			
26.	Plan showing the Aerodrome property limits?			
27.	Information on the Aerodrome ownership?			
Part 3: Information to be communicated to AIS				
28.	Aerodrome name;			
29.	Location of the aerodrome (town, city etc)			



Item	Requirement	Yes	No	Reference Pg. and Comments
30.	Aerodrome Reference Point 1984 (WGS-84);			
31.	Elevation of the highest point of the landing area			
32.	Altitude of each runway end, any significant high and low points along the runway and altitude of the highest point in the touchdown zone area for precision approach runway			
33.	Aerodrome reference temperature			
34.	Details on the Aerodrome non-directional beacon;			
35.	Name, address and phone number(s). of the Aerodrome operator			
36.	Physical characteristics of the Aerodrome			
37.	Runway(s) — true heading, identification number,			
38.	Runway(s) length, width, location of displaced threshold, slope, type of surface,?			
39.	Runway pavement bearing strength (ACN-PCN)			
40.	Type of runway, and for a precision approach runway is there an obstacle free zone			
41.	The length, width and type of surface for runway strips, runway end safety area (RESA) and stopway?			
42.	Taxiway(s) - Length, width and type of surface?			
43.	Co-ordinates of appropriate taxiway centreline points			
44.	Taxiway pavement bearing strength (ACN-PCN)			
45.	Type of surface for apron and aircraft stands?			
46.	Apron pavement bearing strength (ACN-PCN)			
47.	Length of clearway and ground profile?			
48.	Type of approach lighting and visual approach slope indicator,			
49.	Type of Runway marking and lighting			
50.	Type of taxiway marking and lighting			
51.	Type of apron marking and lighting			



Item	Requirement	Yes	No	Reference Pg. and Comments
52.	Visual aids for surface movement guidance and control on taxiway and apron including runway holding point, intermediate holding point, stop bars also location and type of docking visual guidance system			
53.	Secondary power supply			
54.	Location and frequency for VOR check points?			
55.	Location and designation of standard taxi routes?			
56.	WGS coordinates for each aircraft parking stand?			
57.	The geographical coordinates and altitude of any significant obstacles located in the approach and take-off path and around the Aerodrome?			
58.	Aerodrome Obstacle Chart Type A details as applicable			
59.	Location and altitude of the altimeter checkpoint(s)			
60.	Declared distances: TORA, TODA, ASDA and LDA?			
61.	The Disabled Aircraft Removal Plan:			
62.	RFFS protection level, type and quantity of extinguishing agents available at the aerodrome?			
63.	Waivers previously granted by IACM. For each waiver: a) Its reference? b) The issuing authority? c) Its issuing date? d) The date it came into effect? e) Object of the waiver and associated procedures? f) The regulation on which the waiver is based?			
Part 4: Particulars of the aerodrome				
4.1: Aerodrome reporting				
64.	Aerodrome procedures for information collection, recording, sharing and transmission;			



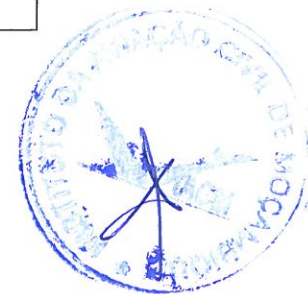
Item	Requirement	Yes	No	Reference Pg. and Comments
65.	Identification and handling of information to be published in the AIP			
66.	Integrity of data during transmission			
67.	Identity of persons (by their names) to whom information is to be communicated			
68.	Transmission format			
69.	Assurance of coordination with the IACM			
70.	Details of procedure for reporting any changes that may affect aircraft operations to AIS and local air traffic services during normal hours of operation			
71.	Details of procedure for reporting any changes that may affect aircraft operations to AIS and local air traffic services outside normal hours of operation			
72.	Procedure for issuance of NOTAM			
4.2 Access to the movement areas				
73.	Aerodrome operator's procedures for control of the movement area access points			
74.	Aerodrome operator's procedures for control and maintenance of fences and gates			
75.	Identity of service & persons responsible (by their name) and telephone numbers H24?			
76.	Coordination with other stakeholders involved (airlines, ground handlers, etc.)			
77.	A plan of aerodrome fences, types			
78.	Wildlife management as far as fences are concerned			
79.	Procedure for recording incidences			
4.3 Aerodrome Emergency Plan (AEP)				
80.	The various type of emergency situations			
81.	Reference to the frequency and nature of emergency exercises			
82.	Date of last exercise			
83.	List of "on the Aerodrome" and "off the Aerodrome" organizations involved in the AEP			



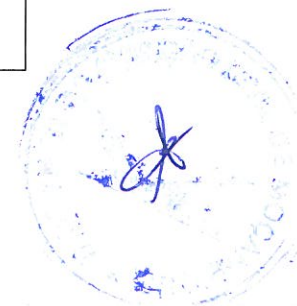
Item	Requirement	Yes	No	Reference Pg. and Comments
84.	Description of Aerodrome Emergency Committee. Roles and responsibilities defined. How often they meet. Date of last meeting.			
85.	Identity of on-scene commander for each type of emergency			
86.	Person responsible for activation of emergency response. Identify by name and H24 contacts.			
87.	AEP development coordination			
88.	Arrangements for regular updating (and distributed to the involved agencies) of the AEP			
89.	Agreements with responding agencies			
90.	Description of the Emergency Operations Centre (EOC); location, accessibility and procedures for activation			
91.	Availability of a Grid Map and procedure for its updating			
92.	Availability of map of the aerodrome vicinity area and procedures for its updating			
4. 4 Rescue and Fire Fighting Services (RFFS)				
93.	RFFS Category provided at the aerodrome			
94.	Description of major firefighting tenders provided to meet the RFFS coverage			
95.	Procedure for reporting of changes of coverage			
96.	Quantities of water in trucks and in reserve			
97.	Quantities and type of foam in trucks and in reserve			
98.	Location and particulars of water hydrants			
99.	Procedure for regular testing of vehicles, tools, facilities etc.			
100.	Number and particulars of personnel per shift			
101.	Availability and accessibility of breathing apparatus			
102.	Information related to operational procedures			
103.	RFFS Station – location and design facilities provided			



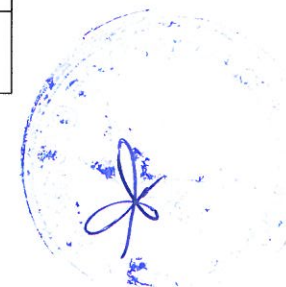
Item	Requirement	Yes	No	Reference Pg. and Comments
104.	Information on RFFS personnel training (needs assessment, programme, plan, Implementation & records)			
105.	Description of procedure for recording of incidences and other vital information			
106.	Availability of a watch tower			
107.	Description of the communication facilities in the watch tower			
108.	Procedure for testing of communication facilities			
109.	Procedure for testing "response time" under different times and conditions			
110.	Description of specialist rescue service if aerodrome is located close to a body of water, swamp or other difficult terrain.			
111.	Description of other tasks the RFFS personnel are allocated: (runway inspections, bird control, etc.)?			
4.5 Inspection of Movement Areas and Control of Obstacle Limitation Surfaces				
112.	Description of the daily inspection procedures (including frequency and timing) of the movement area			
113.	Description of the procedures for reporting, analysis and follow up			
114.	References to third party agreement if the inspections are done by a third party			
115.	The communication facilities and procedures during the inspections			
116.	Name, phone number H24 of the persons responsible for the inspections			
117.	Procedures for evaluation of parts of runway surfaces that may be slippery and measurement of runway friction characteristics			
118.	Procedures for measuring and reporting standing water depth on the runway			
4.6 Aerodrome Visual Aids and Electrical System				
119.	Description of the aerodrome electrical supply and distribution system			
120.	Names (and roles) of personnel responsible for inspection and maintenance of the airfield lighting system during and after normal working hours			



Item	Requirement	Yes	No	Reference Pg. and Comments
121.	Description of the procedures for inspection and maintenance of the airfield lighting (including obstacle lighting), the signs, the marking, the electrical circuits and of the PAPI including			
122.	Procedures for night inspections			
123.	Description of procedures for inspections during and outside Aerodrome normal operating hours			
124.	Description of the inspection reporting system including analysis and follow up.			
125.	Description of procedures for carrying out routine and emergency maintenance			
126.	Description of procedures for inspection after carrying out routine and emergency maintenance			
127.	Description of the secondary power supply system			
128.	Description of the procedures for testing the secondary power supply system			
129.	Description of the procedures for carrying out routine and emergency maintenance of the secondary power supply system			
130.	Description of the procedures for testing the secondary power supply system after carrying out routine and emergency maintenance			
131.	Special procedures applicable during periods of low visibility			
132.	Procedures for reporting of un-serviceability of facilities and/or circuits			
133.	Procedures for flight checking and/or surveying of instrument or lighting by trained technicians after some maintenance tasks			
134.	Procedures for inspection of obstacle lighting			
4.7 Protection of navigation aids (NAVAIDS) located within the aerodrome				
135.	Description of procedures for the protection of radar and NAVAIDS located on the aerodrome to ensure that their performance will not be degraded			
136.	Description of procedures for ground and other maintenance around radar and NAVAIDS located on the aerodrome to ensure that their performance will not be degraded			



Item	Requirement	Yes	No	Reference Pg. and Comments
137.	Procedures for warning and protection from hazardous microwave radiation to personnel working around these installations			
4.8 Maintenance of the Movement Areas				
138.	Procedure for coordination of the planning of maintenance activities in the movement area with IACM			
139.	Description of the procedures for routine maintenance for various facilities and infrastructure of the movement area			
140.	Description of the procedures for periodic maintenance for various facilities and infrastructure of the movement area			
141.	Description of the procedures for major maintenance for various facilities and infrastructure of the movement area			
142.	Procedures for marking and/or lighting of infrastructure around the maintenance work area			
143.	Procedures for temporary closure of infrastructure around the maintenance work area			
144.	Procedures for recording, reporting and evaluating of maintenance related incidences			
4.9 Aerodrome Work Safety				
145.	Description of the procedures for safety assurance during the construction and maintenance works on or close to the movement area			
146.	Procedures for coordination with aerodrome users and IACM of the work scope, scheduling and implementation plan			
147.	Procedures for coordinating with aerodrome users and with IACM instituting changes to the schedule and implementation plan			
148.	Description of the procedures for closures of part of the movement area during the construction and maintenance works			



Item	Requirement	Yes	No	Reference Pg. and Comments
149.	Description of the procedures for communicating information about the project, from the planning stage to the end, to IACM (DAN) including the ATS services			
150.	Procedures to ensure airside safety during the performance of work on the airside during planning stage of the project			
151.	Procedures to ensure airside safety during the performance of work on the airside during the marking, installation of lighting and the cleaning of the working area			
152.	Procedures to ensure airside safety during the performance of work on the airside during commencement of work on the airside			
153.	Procedures to ensure airside safety during the performance of work on the airside during execution of the project			
154.	Procedures to ensure airside safety during the performance of work on the airside during commissioning and re-opening			
155.	Procedures for recording, reporting and analysis of incidences during works implementation			
156.	Procedures publication of a NOTAM prior to commencement of works			
4.10: Apron Management				
157.	Description of arrangements between air traffic control and the apron management unit			
158.	Arrangements for effective and efficient communication between ATC and Apron Management Unit			
159.	Coordination for allocating aircraft parking positions			
160.	Procedure for initiating engine start and ensuring clearance of aircraft push-back			
161.	Aircraft Marshalling service			
162.	Lead (van) service			
4.11 Apron Safety Management				
163.	Assignment of an apron safety officer describing functions and responsibilities			



Item	Requirement	Yes	No	Reference Pg. and Comments
164.	Procedures related to protection from jet blasts, standardized circulation corridors for vehicles and persons and or provision of jet blast fences			
165.	Procedures related to protection during aircraft refuelling operations			
166.	Description of process for handling apron contamination (fuel spillage)			
167.	Procedures for apron sweeping and cleaning			
168.	Apron accident/incident reporting system			
169.	Description of management of safety assurance by contractors, sub-contractors and concessionaires			
4.12: Airside Vehicle Movement Control				
170.	Description of procedures for the control of movement of vehicles on the airside			
171.	Regulations regarding registration and driving of vehicle on the airside			
172.	Regulations regarding communication equipment and other equipment (rotating beacon, color of the vehicle) on the vehicle driving on the airside			
173.	Procedure for issuance and control of Aerodrome vehicle operator permits			
174.	Procedures regarding training and certification of persons driving a vehicle on movement area (apron and maneuvering area)			
175.	Procedures regarding communications with ATS			
176.	A vehicle circulation plan on the apron(s)			
177.	Procedure to be followed by contractors with temporary duties on the airside			
4.13: Wildlife Hazard Management				
178.	Description of the measures to reduce the wildlife hazard			
179.	Identification of the staff allocated and facilities provided for the task of wildlife hazard management			
180.	Description of the wildlife strike recording and reporting system			
181.	Procedure for on-time communication with ATS on observation of presence of wildlife in the movement areas			



Item	Requirement	Yes	No	Reference Pg. and Comments
182.	Procedure for reporting of bird strikes to IACM for onward transmission to IBIS			
4.14: Obstacle Control				
183.	Description and particulars the aerodromes OLS			
184.	Description of the procedure for the control and the notification to IACM of obstacles both inside and outside the aerodrome.			
185.	Description of the coordination of the publication of existence of any obstacles inside and outside the aerodrome in the appropriate aeronautical publications			
186.	Description of the procedure for the coordination with other local authorities, government agencies and private developers inside and outside the aerodrome for the control of obstacles			
187.	Procedures for marking and installation of obstacle lighting where required.			
188.	Procedures for development and updating of the Obstacle Chart(s)			
4.15: Disabled Aircraft Removal Plan				
189.	Description of the roles of the Aerodrome Operator, the holder of the Aircraft Certificate of Registration procedure for the removal of a disabled aircraft on or near the movement area			
190.	Communication with IACM and the Accident Investigation Authority			
191.	Procedure for protection of evidence			
192.	Arrangements with third party for the removal of disabled aircraft			
193.	Procedures for coordination with ATC on the removal process			
194.	Procedures for coordination with ATC on the operation of the aerodrome			
195.	The names, role and telephone numbers of persons responsible for arranging for the removal of disabled aircraft			
196.	Procedure for recording of the removal process			
4.16: Handling of Hazardous Materials				



Item	Requirement	Yes	No	Reference Pg. and Comments
197.	<i>Notes: 1 Hazardous material include explosives, flammable liquids and solids, corrosive liquids, compressed gases, and magnetised or radioactive materials. Hazardous material does not include materials classed by ICAO/IATA as dangerous goods, where freight forwarders and airlines have responsibilities for safe packaging and handling procedures</i> <i>2 The arrangements to deal with an accidental spillage of hazardous materials are to be set out in the aerodrome emergency plan.</i>			
198.	Description of the procedures for and facility for handling and storage of hazardous materials			
199.	Identification of an isolated parking stand for aircraft in-transit and known to be transporting hazardous materials			
200.	Procedures for informing the Rescue and Firefighting Service about an aircraft with dangerous goods on board.			
4.17: Low Visibility Operations				
201.	Description of the procedures for low-visibility operations, including the measurement and reporting of runway visual range (RVR) as and when required and its transmission to the ATS			
202.	Procedure for coordination with ATS for inspection of maneuvering area when under low-visibility			
203.	Procedure to check visual aids lighting intensity to adjust brilliance as appropriate			
204.	Procedures for runway inspections during low visibility periods			
205.	Procedure to keep RFFS on stand-by			
206.	Procedure for informing, in real time, all personnel involved in airside operations that low-visibility procedures have been activated			
207.	Procedures to restrict vehicle movement during low visibility operations			
208.	Procedures for recording, reporting, analysis and following up of low visibility operationally related incidents and accidents			
4.18: Aerodrome Administration				



Item	Requirement	Yes	No	Reference Pg. and Comments
209.	Is the Aerodrome Manual properly amended and current ?			
210.	Have all the amendments been reported (on time) to IACM?			
211.	Have the amendments been transmitted to ALL Aerodrome manual holders?			
212.	The Aerodrome manual includes the current Aerodrome Organizational Chart and names and titles of key personnel with their main responsibilities			
213.	Does it specifically show the Name, titles and contact details of the person with overall responsibility for Aerodrome Safety (during normal and off operational hours)?			
214.	Description of the various Aerodrome Committees (membership, roles and dates of last meetings)			
215.	Does AM include a MOZ CATS 139, Vol. I Compliance Checklist?			
216.	Does the AM include confirmation to compliance with any condition(s) attached with the previous certificate?			
PART 5 : SAFETY MANAGEMENT SYSTEM (SMS)				
217.	Does the AM include a description of the SMS the Aerodrome is implementing?			
218.	Does the AM include a description of the approved Aerodrome Safety Policy?			
219.	Does the AM indicate who is the Accountable Executive as far as SMS implementation is concerned?			
220.	Names and contact details of the Aerodrome Safety Officer (ASO) with responsibility for SMS implementation.			
221.	Does ASO report to Aerodrome Manager?			
222.	Does the AM include an SMS Implementation Plan or Schedule?			
223.	Does the AM include an assessment of the progress of implementation of SMS			



This completed Compliance Checklist ofAerodrome Manual has been approved for use by:

NAME OF TEAM LEADER	_____		
MOZAMBIQUE CIVIL AVIATION AUTHORITY	FULL NAME IN BLOCK LETTERS	DATE	AMENDMENT NUMBER

Approved by:

SIGNATURE OF DIRECTOR OF AIR NAVIGATION	FULL NAME IN BLOCK LETTERS	DATE	AMENDMENT NUMBER

Date: 17/03/2016

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Appendix 1 – CT 139-001



AERODROME MANUAL TEMPLATE

1. Airport Operator Statement and Aerodrome Manual Approval

Airport's Name:					
Airport Owner's Name:	Airport Operator's Name:				
Latitude:	Longitude:				
Airport Manager's Name:					
Airport Certificate Number:	Date of Issuance (dd/mm/yyyy):				
STATEMENT - AIRPORT OPERATOR					
I hereby certify that the information in this Aerodrome Manual is correct and that no relevant information has been omitted. I accept and will comply with all the specifications contained herein.					
<table border="1" style="width: 100%;"> <tr> <td style="width: 60%; height: 50px;"></td> <td style="width: 40%; height: 50px;"></td> </tr> <tr> <td>Name and Signature of the Airport Operator</td> <td>Date</td> </tr> </table>				Name and Signature of the Airport Operator	Date
Name and Signature of the Airport Operator	Date				
APPROVAL					
This Aerodrome Manual is approved.					
<table border="1" style="width: 100%;"> <tr> <td style="width: 60%; height: 80px;"></td> <td style="width: 40%; height: 80px;"></td> </tr> <tr> <td>President and Chief Executive Officer, Instituto de Aviação Civil de Moçambique</td> <td>Date</td> </tr> </table>				President and Chief Executive Officer, Instituto de Aviação Civil de Moçambique	Date
President and Chief Executive Officer, Instituto de Aviação Civil de Moçambique	Date				

2. Distribution List

Manual	Identification Number	Holder's Name
	1	
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3. Record of Amendments

Nº	DATE RECEIVED	DATE INSERTED	INSERTED BY	SIGNATURE
1				
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4				
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Amendments to the Aerodrome Manual will be done in accordance with the procedure described at Appendix 1, Part 4 of the “Airport Certification Guide” and by taking out the pages to be replaced and by inserting the new ones.

4. Exemptions Record

System for recording and publishing Exemptions				
Nr	Certificate Nr.	Issue of exemption	Deadline	Obs.

5. List of Pages in Effect

List of pages in Effect											
Number: 1 List Pages in effect						Date: May 02, 2018					
Front Page		Foreword		Approval		Distribution List		Amend. Record		Exemptions Record	
Page	Revision	Page	Revision	Page	Revision	Page	Revision	Page	Revision	Page	Revision
0	0	i	0	ii	0	iii	0	iv	0	v	0
List Pages in effect		Table of contents		General		Chapter 1		Chapter 2		Chapter 3	
Page	Revision	Page	Revision	Page	Revision	Page	Revision	Page	Revision	Page	Revision
vi	0	vii	0	10	0	1-1	0	2-1	0	3-1	0
		viii	0	11	0	1-2	0			3-2	0
		ix	0	12	0	1-3	0			3-3	0
				13	0	1-4	0			3-4	0
				14	0	1-5	0			3-5	0
										3-6	0
										3-7	0
Chapter 4		Chapter 5		Chapter 6		Chapter 7		Chapter 8			
Page	Revision	Page	Revision	Page	Revision	Page	Revision	Page	Revision		
4-1	0	5-1	0	6-1	0		0		0		
4-2	0			6-2	0						
4-3	0			6-3	0						
4-4	0			6-4	0						
4-5	0			6-5	0						
4-6	0			6-6	0						
4-7	0			6-7	0						
4-8	0			6-8	0						
4-9	0			6-9	0						
4-10	0			6-10	0						
4-11	0										
4-12	0										
4-13	0										

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Chapter 1 - GENERAL

1.1. OBJECT AND SCOPE OF THE AERODROME MANUAL

This Aerodrome Manual (AM) has been prepared because it is a condition for Airport certification and is part of the Airport certificate. The principles for the certification of Airport operators are defined in the applicable national regulations.

This Aerodrome Manual formulates the standards that will be respected, the operational procedures in effect at the (name of the airport) International Airport as they exist at the time of delivery of the Airport certificate or as modified from time to time. It may be used as:

- i) Legal source of reference, between the Airport operator and the IACM relative to Airport standards, conditions and levels of service to be respected in order to maintain the Airport operating certificate valid;
- ii) Reference document for Airport inspections;
- iii) Reference document for Airport users;
- iv) Legal registration document pertaining to any modification or approved deviation from Airport standards and the conditions and levels of service pertaining to airside activities.

1.2. LAW AND REGULATIONS REQUIREMENTS

This Aerodrome Manual is published according with the requirements of the Mozambique national regulations and other applicable regulations.

The installations, services and information describe in the AM are in accordance with the standards described in:

- i) MOZ-CAR 139 and MOZ-CATS 139, Volume I: or
- ii) As indicated an approved deviation from MOZ-CATS 139, Volume I.

1.3. AIRPORT OPERATING CONDITIONS

The (name of the airport) is opened 24 hours a day, 365 days a year. It can be used under VFR and IFR conditions. It is equipped with a non-directional beacon (NOB) and with an omnidirectional beacon coupled with distance measuring equipment (VORIDME) and approaches RNAV DME for all runways.

Unless the Airport is closed or cannot be used in whole or in part, it will be usable by all under the same conditions.

1.4. AERONAUTICAL INFORMATION SYSTEM AND PUBLISHING PROCEDURES

To describe in details the aeronautical information process and the publishing procedures used at the airport taking into account the established quality system procedures.

1.5. AIRCRAFT MOVEMENTS RECORDING SYSTEM

To describe the actual system used for recording the aircraft movements at the airport.

1.6. AIRPORT OPERATOR'S OBLIGATION

i) The Airport operator must:

- a) Conform to standards outlined in Airport standards and recommended practices publications, as they exist at the time of delivery of the Airport certificate, and such other conditions specified by the IACM on the Airport certificate;
- b) Free of charge, upon a IACM inspector, allow access to Airport facilities and provide him or her with the equipment required to carry

out the Airport inspection;

- c) Check, upon receipt, each new edition of the Aeronautical Information Publications, and immediately inform the IACM after review, of any inaccuracies contained in the information relative to the Airport in question;
- d) Inform the IACM, in writing, at least 14 days before the occurrence at the aerodrome, its facilities or services, of any change that would invalidate information contained in the Aeronautical Information Publications;
- e) Whenever circumstances so indicate in order to maintain safety of operations inspect the Airport:
 - i. As soon as feasibly possible after an aircraft incident or accident;
 - ii. During such times that the Airport or its designated facilities are under construction or repairs;
 - iii. As soon as any situation arises that could jeopardize air navigation safety.
- f) Except as provided under item d), notify the IACM in writing of any change in Airport operations within 14 days following the date that such change took place.
 - ii) The Airport operator must immediately notify the IACM, and ensure that the pertinent ATS section is notified, as soon as he learns about the following occurrences taking place;
- g) Any object penetrating an obstacle limitation surface;
- h) The presence of obstacles or the existence of a dangerous situation jeopardizing aviation safety at the Airport or adjacent to it;
- i) A decrease in the level of services provided at the Airport and as specified in the Aeronautical Information Publication;
- j) Closure of part of the Airport maneuvering area;

- k) Any other situation that could jeopardize Airport safety and for which certain preventive measures should justifiably be taken.

The Airport operator may remove from the surface any vehicle or obstacle that could potentially jeopardize aviation safety at the Airport or adjacent to it.

1.7. USE OF COMMON REFERENCE SYSTEMS

- i) The World Geodetic System – 1984 (WGS-84) is used as the horizontal reference system to express aeronautical geographical coordinates for aerodromes.
- ii) The Mean Sea Level datum is used as the vertical reference system (elevation) at aerodromes.
- iii) The Gregorian calendar and Coordinated Universal Time is used as the temporal reference system.

1.8. DEFINITIONS AND ACRONYMS

1.8.1. DEFINITIONS

It may be adopted and not limited to the definitions below and they should be in accordance with the latest amendments of Annex 14 and MOZCATS 139.

Airport certificate. A certificate issued by the appropriate authority under applicable regulations for the operation of an aerodrome.

Aerodrome Manual. Manual which is part of the Airport certification request and part of the Airport certificate.

Airport operator. Concerning a certified airport, means the holder of the Airport certificate.

Apron. A defined area, on a land aerodrome, intended to accommodate aircraft for purposes of loading or unloading passengers, mail or cargo, fuelling, parking or maintenance.

Certified Airport. An Airport whose operator has been granted an Airport

.

Installations and Airport equipment. Installations and equipment, inside or outside Airport boundaries, which have been erected or installed and maintained for the take-off and landing of aircraft and their surface movements.

Maneuvering area. That part of an Airport to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

Marker. An object displayed above ground level in order to indicate an obstacle or delineate a boundary.

Marking. A symbol or group of symbols displayed on the surface of the movement area in order to convey aeronautical information.

Movement area. That part of an Airport to be used for the take-off, landing and taxiing of aircraft, consisting of the maneuvering area and the apron(s).

Obstacle. All fixed (whether temporary or permanent) and mobile objects, or parts thereof, that:

- a) Are located on an area intended for the surface movement of aircraft; or
- b) Extend above a defined surface intended to protect aircraft in flight; or
- c) Stand outside those defined surfaces and that have been assessed as being a hazard to air navigation.

Obstacle Free Zone (OFZ). The airspace above the inner approach surface, inner transitional surfaces, and balked landing surface and that portion of the strip bounded by these surfaces, which is not penetrated by any fixed obstacle other than a low-mass and frangibly mounted one required for air navigation purposes.

Obstacle limitation surface. Various surfaces which define the volume of air space to be kept free of obstacles at an Airport and in its vicinity in order to permit aircraft safe movements.

Runway strip. A defined area including the runway and stopway, if provided, intended:

- i) To reduce the risk of damage to aircraft running off a runway; and

- ii) To protect aircraft flying over it during take-off or landing operations.

Safety Management System (SMS). A systemic approach to managing safety including the necessary organizational structure, accountabilities, policies and procedures.

Standards and Practices: Any references to standards and recommended practices in this document is about the standards and recommended practices (SARPs) of MOZ-CATS 139, Volume I, Civil Aviation Convention and to national regulations and practices as may be amended from time to time.

Taxiway strip. An area including a taxiway intended to protect an aircraft operating on the taxiway and to reduce the risk of damage to an aircraft accidentally running off the taxiway.

Unusable Area. Part of the movement area not available for aircraft movement.

Work Area. Part of an Airport where maintenance or construction works are taking place.

(To insert other appropriate definitions by alphabetical order).

1.8.2. ACRONYMS

AIP.	Aeronautical Information Publications
AIS.	Aeronautical Information Service
AM.	Aerodrome Manual
AVOP.	Airport Vehicle Operator Permit
IACM.	Instituto de Aviação Civil de Moçambique
ICAO.	International Civil Aviation Organization
NOTAM.	Notice to Airmen
SMS.	Safety Management System
SMSM.	Safety Management System Manual

Chapter 2 - INFORMATION ON AIRPORT LOCATION

2.1. AIRPORT PLANS

The following plans will be found in Appendices:

- i) Airport Plan No. ????, at scale, showing the main operational installations, such as wind direction indicators. (See Appendix ???).
- ii) Plan No. ????, at scale, showing the different areas of the Airport and the delimitation between the apron and the maneuvering area, the cargo area, the terminal building and areas reserved for navigational aids and meteorological installations. (See Appendix ???).
- iii) Plan No. ????, showing the apron area with aircraft stands marking. (See Appendix ???)
- iv) Plan No. ????, at scale, showing the maneuvering area. (See Appendix ???).
- v) Plan No. ????, showing the lighting system, the PAPI, the various signs (direction, location, mandatory signs, etc.), (See Appendix ???)
- vi) Plan No. ????, showing the Airport boundaries (See Appendix ???).
- vii) Information on the legal ownership of the aerodrome.

N.B.: Plans shall be presented under paper format and annexed to the Aerodrome Manual.

Plans shall also be available under electronic format and made available to the IACM.

Chapter 3 - AIRPORT INSTALLATIONS

3.1. AIRPORT DATA

Airport name	
Location of the aerodrome	
Geographical Coordinates (reference point)	
Airport altitude	
Threshold altitude	
Threshold altitude	
Runway end altitude	
Runway end altitude	
Altitude of any significant high point	
Highest point of the TDZ, (precision rwy).	
Airport Reference Temperature	
Details on Airport beacon	<i>(location and type (flashing or rotating))</i>
Name of the Airport operator Address	
Address of the Airport operator	
Airport operator telephone numbers	
Name of the Accountable executive	

3.2. AIRPORT DIMENSION DATA

General Information			
Runways			
Identification numbers	07	25	13 31
Runway code			
True orientation			
Length			
Width			
Displaced threshold location			
Slope			
Type of surface			
Type of runway	(precision, non-prec)		
Type of Critical Aircraft			

Clearances		
For precision approach rwy to indicate obstacle free area		
Runway strip - lenght		
Runway strip - width		
Runway strip - type of surface		
RESA - lenght		
RESA - width		
RESA - type of surface		
Stopway - lenght		
Stopway - width		
Stopway - type of surface		

Taxiways		
Identification of taxiways		
Taxiway - length		
Taxiway - width		
Taxiway - type of surface		

Aprons		
Apron Identification		
Apron - length		
Apron - width		
Apron - type of surface		

Stopways		
Runway Identification		
Stopway - length		
Stopway - ground profil		

Lighting		
Runways Identification		
Approach lighting type		
PAPI/VASIS		
Runway Lights		
Windsock (Location, Quantity and Illuminated)		

Taxiways Identification		
Taxiway lighting		
Apron Identification		
Apron lighting		

Marking		
Runway Identification		
Runway marking		
Taxiway Identification		
Taxiway marking		
Apron Identification		
Apron marking		

Visual Guidance		
Taxiway identification		
Visual guidance - taxiway		
Runway holding position		
Intermediate holding position		
Stopway lights		

Visual Docking System		
Visual docking system - Location		
Visual docking system - Type		

<i>Auxiliary Power Supply</i>		
Lighting auxiliary power supply		

VOR Checkpoint		
VOR checkpoint - location		
VOR checkpoint - radio frequency		

Ground Circulation Corridors		
Identification and location of ground circulation corridors		

Geographical Coordinates - threshold, TWY, aircraft stands and obstacles		
Geographical Coordinates - Threshold		
Geographical Coordinates - Threshold		
Geographical Coordinates - Threshold		
Geographical Coordinates - TWY C. line		
Geographical Coordinates - TWY C. line		
Geographical Coordinates - TWY C. line		
Geographical Coordinates - TWY C. line		
Geographical Coordinates - Aircraft stand		
Geographical Coordinates - Aircraft stand		
Geographical Coordinates - Aircraft stand		
Geographical Coordinates - Aircraft stand		
Geographical Coordinates - Aircraft stand		
Geographical Coordinates - Aircraft stand		
Obstacles - Geographical Coordinates		
Obstacles - Altitude		
Obstacles - Geographical Coordinates		
Obstacles - Altitude		
Obstacles - Geographical Coordinates		
Obstacles - Altitude		

Pavement		
Pavement - type of surface		
Pavement - bearing strength		
Pavement - type of surface		
Pavement - bearing strength		
Pavement - type of surface		
Pavement - bearing strength		
Pavement - type of surface		
Pavement - bearing strength		

Altimeter Checkpoint		
Altimeter checkpoint - location		
Altimeter checkpoint - altitude		
Altimeter checkpoint - location		
Altimeter checkpoint - altitude		
Altimeter checkpoint - location		
Altimeter checkpoint - altitude		

Declared Distances	07	25	13	31
Declared Distances - TORA				
Declared Distances - TODA				
Declared Distances - ASDA				
Declared Distances - LOA				

Note: Data for, and the method used to calculate, declared distances and elevations at the beginning and end of each declared distance should be shown attached in this manual.

Disabled Aircraft Removal Plan		
DARP ¹ - coordinator name		
DARP - coordinator telephone numbers		
DARP - coordinator telex number		
DARP - coordinator fax number		
DARP - coordinator e-mail		
DARP - available means		

Airport Rescue and Fire Fighting	
AD RFF – category	

¹ DARP = Disabled Aircraft Removal Plan
Second Edition. Revision 0, August 16, 2018

Waivers		
Waiver No 1 Reference Number		
Waiver No 1 Issuing authority		
Waiver No 1 Date of issuance		
Waiver No 1 Date of application		
Waiver No 1 Subject		
Waiver No 1 Associated procedures		
Waiver No 1 Regulation reference		
Waiver No 2 Reference Number		
Waiver No 2 Issuing authority		
Waiver No 2 Date of issuance		
Waiver No 2 Date of application		
Waiver No 2 Subject		
Waiver No 2 Associated procedures		
Waiver No 2 Regulation reference		

Chapter 4 - OPERATIONAL PROCEDURES AND AIRPORT SAFETY MEASURES

4.1. AIRPORT REPORTING

Particulars of the procedures for reporting any changes to the Airport information set out in the AIP and procedures for requesting the issue of NOTAMs, including the following:

- I. Arrangements for reporting any changes to the IACM and recording the reporting of changes during and outside the normal hours of Airport operations;
- II. The names and roles of persons responsible for notifying the changes, and their telephone numbers during and outside the normal hours of Airport operations; and
- III. The address and telephone numbers, as provided by the IACM, of the place where changes are to be reported to the IACM.

4.2. ACCESS TO THE AIRPORT MOVEMENT AREA

Particulars of the procedures that have been developed and are to be followed in coordination with the agency responsible for preventing unlawful interference in civil aviation at the Airport and for preventing unauthorized entry of persons, vehicles, equipment, animals or other things into the movement area, including the following:

- i) The role of the Airport operator, the aircraft operator, Airport fixed-base operators, the Airport security entity, the IACM and other government departments, as applicable; and;
- ii) Control and maintenance of the Airport access points;
 - Identification of authorized persons on the airside or movement area.
- iii) Control and maintenance of the Airport perimeter fences;
 - Describe who is responsible for the maintenance of fence and gates.
- iv) The names and roles of the personnel responsible for controlling access to the aerodrome, and the telephone numbers for contacting them during and after working hours;

- v) Coordination with other Airport operators, such as apron managed by a third party, airlines, construction contractors, etc.;
- vi) A plan of the Airport fences shall be attached and include:
 - Describe the fences at the Aerodrome (giving height, location), if different fence heights exist, describe locations;
 - Lighting fences and safety barriers to the all critical points.
- vii) The type of fences shall be described; the access points shall be identified and the means of controlling these access points shall be described for each of them;
- viii) Wildlife management as far as fences are concerned.

4.3. AIRPORT EMERGENCY PLAN

Particulars of the Airport emergency plan, including the following:

- i) Plans for dealing with emergencies occurring at the Airport or in its vicinity, including in sites close to water, swampy areas or difficult terrain, the malfunction of aircraft in flight; structural fires; sabotage, including bomb threats (aircraft or structure); unlawful seizure of aircraft; and incidents on the Airport covering "during the emergency" and "after the emergency" considerations;
- ii) Details of tests for Airport facilities and equipment to be used in emergencies, including the frequency of those tests;
- iii) Details of exercises to test emergency plans, including the frequency of those exercises;
- iv) A list of organizations, agencies and persons of authority, both *on-* and *off-* aerodrome, for site roles; their telephone and facsimile numbers, *e-mail* and SITA addresses and the radio frequencies of their offices;
- v) The establishment of an Airport emergency committee to organize training and other preparations for dealing with emergencies; and
- vi) The appointment of an on-scene commander for the overall emergency operations;
- vii) The Airport operator shall identify the service and the person responsible for managing emergency situation (name of the person) and his/her (their) telephone number(s) 24h/24h;

- viii) The development of the Airport emergency plan has been coordinated with outside agencies such as civil protection, police, hospital, ambulances, air traffic control, airlines, etc.;
- ix) The plan is updated regularly and copies are distributed to participating agencies.

The Airport emergency plan shall cover the following situations:

- a) Aircraft crash on the aerodrome;
- b) Accident crash outside the aerodrome;
- c) Illegal acts (hijacking, bomb threat, anonymous calls etc.);
- d) Building fire;
- e) Fuel spill;
- f) Dangerous goods;
- g) Natural catastrophe (earth quake, storm, etc.);
- h) Bomb threat in a building;
- i) Fire in a building;
- j) Health emergency;
- k) Etc..

4.4. RESCUE AND FIRE-FIGHTING SERVICES

- a) Policy statement on the RFF categories to be provided.
- b) Describe the procedure for with on the continuing basis, that aerodrome operators notify changes in the level of aircraft rescue and fire fighting (RFF) protection normally available at aerodrome to air traffic service (ATS) and AIS.
- c) Where the senior aerodrome fire officer or designated fire watch officers have specific safety accountabilities, these should be included in the relevant chapter of the aerodrome manual.
- d) Policy and procedures indicating how depletion of the RFF service is to be managed. This should include the extent to which operations are to be restricted, how pilots are to be notified and the maximum duration of any depletion.
- e) At aerodromes where a higher category of RFF is available by prior arrangement, the aerodrome manual should clearly state the actions necessary to upgrade the

facility. Where necessary, this should include actions to be taken by other departments.

f) The aerodrome operator's objectives for each RFF category provided should be defined, including a brief description of:

- 1) amounts of extinguishing agents provided;
- 2) discharge rates;
- 3) number of foam-producing appliances;
- 4) manning levels;
- 5) levels of supervision.

g) Procedures for:

- 1) monitoring the aeroplane movement areas for the purpose of alerting RFF personnel;
- 2) indicating how the adequacy of the response time capability of the RFF services throughout their functions and locations is monitored and maintained;
- 3) Indicating how RFF personnel engaged in extraneous duties are managed to ensure that response capability is not affected.

h) Where the aerodrome provides specialist equipment such as rescue craft, emergency tenders, hose layers, and appliances with aerial capability, details should be included in the aerodrome manual. Procedures to be followed if these facilities are temporarily unavailable should also be included.

i) Where the aerodrome is reliant upon other organizations to provide equipment which is essential for ensuring the safe operation of the aerodrome (perhaps water rescue), policies or letters of agreement should be included in the aerodrome manual. Where necessary, contingency plans in the event of non-availability should be described.

j) A statement describing the process by which aerodrome operators ensure the initial and continued competence of their RFF personnel, including the following:

- 1) realistic fuel fire training;
- 2) breathing apparatus training in heat and smoke;
- 3) first aid;
- 4) low visibility procedures (LVP);
- 5) any legal requirements;

- 6) health and safety policy with regard to training of personnel in the use of respiratory protection equipment and personal protection equipment.
- k) Procedures indicating how accidents in the immediate vicinity of the aerodrome are to be accessed. Where difficult environs exist, the aerodrome manual should indicate how these are to be accessed.
- l) Where local authorities or the aerodrome operator expects the RFF facility to respond to domestic fires or special services, procedures for managing their impact upon normal aeroplane RFF responses should be included.
- m) Where the aerodrome operator expects the RFF facility to respond to aeroplane accidents landside, the policy should be clearly described, including procedures to manage the effects on continued aeroplane operations.
- n) The availability of additional water supplies should be described.
- o) Aerodrome operator's arrangements for ensuring the adequacy of responses in abnormal conditions, i.e. LVP.

4.5. INSPECTION OF THE MOVEMENT AREA

Particulars of the procedures for the inspection of the Airport movement area, including the following:

- a) Routine aerodrome inspections, including lighting inspections, and reporting, including the nature and frequency of these inspections.
- b) Inspecting the apron, runways and taxiways following a report of debris on the movement area, an abandoned take-off due to engine, tire or wheel failure, or any incident likely to result in debris being left in a hazardous position.
- c) Sweeping of runways, taxiways and aprons.
- d) Measurement and promulgation of water, slush and other contaminants including depths on runways and taxiways.
- e) A process for determine and providing relevant information that a runway, or part of, may be slippery when wet, including the minimum friction level for reporting of slippery runway conditions and the type of friction measuring device used;
- f) Assessment and promulgation of runway surface conditions:
 - 1) details of inspection intervals and times;
 - 2) completion and effective use of an inspection checklist;

- 3) arrangements and methods for carrying out inspections on FOD, lighting, pavement surface, grassing, etc.;
 - 4) arrangements for reporting the results of inspections and for follow-up;
 - 5) arrangements and means of communication with air traffic control during an inspection;
 - 6) arrangements for keeping an inspection logbook and the location of the logbook.
- i) The names and roles of persons responsible for carrying out inspections, and their telephone numbers during and after working hours;
 - ii) Develop and implement aerodrome inspections programmes in the interests of safety, efficiency and regularity of aircraft operations?

4.6. MAINTENANCE OF THE MOVEMENT AREA

Particulars of the facilities and procedures for the maintenance of the movement area, including:

- a) Arrangements for maintaining the paved areas, including the runway friction assessments;
- b) Arrangements for maintaining the unpaved runways and taxiways;
- c) Arrangements for maintaining the runway and taxiway strips;
- d) Arrangements for the maintenance of Airport drainage;
- e) Arrangements for assess of coverage of water on runway surfaces, taking account of the need for aircraft crews to make operational adjustments in accordance with performance requirements in some flight manuals;
- f) Arrangements for maintaining the visual aids, including the measurement of intensity, beam spread and orientation of lights;
- g) Arrangements for maintaining the obstacle lighting;
- h) Arrangements for reporting and action taken in the event of failure or unsafe occurrence.
 - surface debris and regularity
- i) Arrangements for the safe return of a runway to operational status.

The manual shall also describes the following items concerning the scheduled and preventive maintenance:

- The frequency of the scheduled maintenance activities such as runway, taxiway and apron surfaces, rubber removal, grass cutting, drainage system, etc..
- The methods to evaluate the pavement conditions

4.7. VISUAL AIDS AND AIRPORT ELECTRICAL SYSTEM

Particulars of the procedures for the inspection and maintenance of aeronautical lights (including obstacle lighting), signs, markers and Airport electrical systems, including the following:

- a) Arrangements for carrying out inspections during and outside the normal hours of Airport operation, and the checklist for such inspections;
- b) Arrangements for recording the result of inspections and for taking follow-up action to correct deficiencies;
- c) Arrangements for carrying out routine maintenance and emergency maintenance;
- d) Arrangements for secondary power supplies, if any, and, if applicable, the particulars of any other method of dealing with partial or total system failure;
- e) The names and roles of the persons responsible for the inspection and maintenance of the lighting, and the telephone numbers for contacting those persons during and after working hours;
- f) Procedures for electrical power supply during low visibility operations;
- g) Responsibilities with respect to the aerodrome ground lighting system;
- h) The plan for lighting, signs and markings for individual elements;
- i) A full description of all visual aids available on each approach, runway, taxiway and apron, including signs, markings and signals;
- j) Procedures for operational use and brilliancy settings of the lighting system;
- l) Standby and emergency power arrangements, including operating procedures both in LVP and during main power failure situations;
- m) Procedures for routine inspection and photometric testing of approach lights, runway lights, VASIS and PAPIs;

- n) The location of and responsibility for obstacle lighting on and off the aerodrome;
- o) Procedures for recording inspection and maintenance of visual aids and actions to be taken in the event of failures;
- p) The control of work, including trenching and agricultural activity, which may affect the safety of the aeroplane; and

4.8. AIRPORT WORKS SAFETY

Particulars of the procedures for planning and carrying out construction and maintenance work safely (including work that may have to be carried out at short notice) on or in the vicinity of the movement area which may extend above an obstacle limitation surface, including the following:

- a) Arrangements for communicating with air traffic control during the progress of such work;
- b) The names, telephone numbers and roles of the persons and organizations responsible for planning and carrying out the work, and arrangements for contacting those persons and organizations at all times;
- c) The names and telephone numbers, during and after working hours, of the Airport fixed-base operators, ground handling agents and aircraft operators who are to be notified of the work;
- d) A distribution list of work plans, if required.

This section describes the procedures established by the Airport operator to insure the work security from the planning stage and including the beginning of work, the work itself, the fencing around the work area (lighting, marking, cleaning of the work area and around the work area, inspection of the work area once the work is completed).

Procedures related to closure of various areas of the movement area will be mentioned.

The manual shall also deal with information concerning coordination and information to be provided to those affected by the work (airlines, ATS, etc.). It is important to note that the coordination shall be done from the planning stage of the project, it may means months if not year in advance.

The control and operational procedures are evaluated on a case by case basis with the Air Traffic Control, keeping in mind the safety of aircraft.

At all times, the Air Traffic Control will be informed of the work schedule and the nature of work to be done.

Such work having a significant impact on the level of service on the air safety will be published in the appropriate aeronautical publication.

Procedure for evaluating the impact of the change on the safety of the existing operation when a change to the aerodrome physical characteristics, facilities or equipment is proposed.

For construction work on the air side, a "Plan Construction Operation (P.C.O.)" will be prepared for each project. The following items, but not exclusively, shall be included:

- i. An escort service shall be provided by the Airport operator. The persons doing the escort shall be the holder of the required permits such as permit to operate a vehicle on the airside including the maneuvering area.
- ii. Meetings between the Airport operator, the Air Traffic Control and the person responsible for the project will take place at the beginning of the work and regularly thereafter. The main purpose of these meetings being to coordinate the implementation of the project and to establish operational procedures to maintain the air safety.
- iii. The manual shall include all the details related to the work: preventive measures, signage, markings, etc.;
- iv. Measures will be taken in order to isolate the work area from the operational area using lighting, signage, markers, etc... These measures could be evaluated by the IACM Airport safety division if need be.
- v. Procedures for coordinating with aerodrome users and with IACM instituting changes to the schedule and implementation plan.
- vi. Description of the procedures for communicating information about the project, from the planning stage to the end, to IACM (DAN) including the ATS services.

4.9. APRON MANAGEMENT

A circulation plan on the apron shall be attached in appendix.

Particulars of the apron management procedures, including the following:

- i) Arrangements between the air traffic control and the apron management unit;
- ii) Arrangements for allocating aircraft parking positions;
- iii) Arrangement for visual monitoring of aircraft stand clearances and the control of vehicle movement on aprons;
- iv) Arrangements for initiating engine start and ensuring clearance of aircraft push-back;
- v) Provide radiotelephony communications facilities to apron management services;
- vi) Marshaling service;
- vii) Leader (van) service.

The Aerodrome Manual shall provide information concerning the physical means, the human resources and apron management procedures including, amongst others:

- i) The development of a parking plan;
- ii) The identification of support equipment parking area;
- iii) For allocating aircraft parking spots (planning, communication to interested parties, compatibility with each aircraft stand, etc.);
- iv) For aircraft marshaling to the aircraft stand and the use of the aircraft docking system, if applicable;

The plan shall clearly showing the limits between the apron and the maneuvering area included in appendix to the AM

4.10. APRON SAFETY MANAGEMENT

Procedures to ensure apron safety, including:

- i) Protection from jet blasts;
- ii) Enforcement of safety precautions during aircraft refuelling operations;
- iii) Fuel spills;

- iv) Apron sweeping and cleaning;
- v) Arrangements for reporting incidents and accidents on the apron;
- vi) Arrangements for assessing the safety compliance of all personnel working on the apron
- vii) Arrangements for the removal of foreign objects;
- viii) Protection measures to ensure the safety of personnel working on the airside, including the establishment of corridors for vehicles circulation, wearing safety jackets, etc.

This section shall also include information on procedures to be followed by third parties working on the airside, amongst others:

- a) The specifications to be imposed on third parties;
- b) The procedures to make sure that the above-mentioned specifications are followed and that measures are taken by the Airport operator if it is not the case.

4.11. AIRSIDE VEHICLE CONTROL

Particulars of the procedure for the control of surface vehicles operating on or in the vicinity of the movement area, including the following:

- i) Details of the applicable traffic rules (including speed limits and the means of enforcing the rules); and

The Aerodrome Manual shall also makes reference to:

- i) Airport circulation rules to ensure aircraft safety;
- ii) Procedures for deliverance of «Airside Vehicle Operator Permit (AVOP)», radio communication procedures, driving under low visibility conditions, knowledge of the critical and sensitive areas;
- iii) Procedures for communicating with the Air Traffic Control (phraseology, wording, reporting points etc.);
- iv) Procedures to ensure that vehicles proceeding on the airside are properly equipped (radio communication for circulation on maneuvering area, vehicle color, beacon lights, etc.;

This section shall also include information on procedures to be followed by third parties working on the airside, amongst others:

- a) The specifications to be imposed on third parties;
- b) The procedures to make sure that the above-mentioned specifications are followed and that measures are taken by the Airport operator if it is not the case.

4.12. WILDLIFE HAZARD MANAGEMENT

Particulars of the procedures to deal with the danger posed to aircraft operations by the presence of birds or mammals in the Airport flight pattern or on movement area, including the following:

- a) Arrangements for assessing wildlife hazards;
- b) Arrangements for implementing wildlife control programs;
- c) Arrangements for collect and forward wildlife strike reports to IACM; and
- d) The names and roles of the persons responsible for dealing with wildlife hazards, and their telephone numbers during and after working hours.

In this section, we shall also mention:

- a) The availability hours of the wildlife control service;
- b) The means available for wildlife control;
- c) The process to mitigate against an increase or potential increase in the wildlife strike hazard due to land-use development likely to attract wildlife around an aerodrome

The certification requirements for wildlife control staff and the follow-up of these certifications shall be mentioned.

If wildlife control is done by a third party, the terms and conditions of the agreement shall be stated.

Local procedures shall be clearly identified (grass cutting, migratory animals, migratory birds, etc.). The local procedures shall adapt to changing conditions.

4.13. OBSTACLE CONTROL

Particulars setting out the procedures for:

- a) Monitoring the obstacle limitation surfaces and Type A Chart for obstacles in the take-off surface;
- b) Controlling obstacles within the authority of the Airport operator;
- c) Monitoring the height of buildings or structures within the boundaries of the obstacle limitation surfaces;
- d) Controlling new developments in the vicinity of the aerodrome;
- e) Notifying the IACM of the nature and location of obstacles and any subsequent addition or removal of obstacles for action as necessary, including amendment of the AIS (AIP).

This section shall also describe or include:

- i) The Airport zoning regulations, if any;
- ii) The monitoring of obstacle limitation surfaces on working site (construction projects);

4.14. REMOVAL OF DISABLED AIRCRAFT

Particulars of the procedures for removing a disabled aircraft on or adjacent to the movement area, including the following:

- a) The roles of the Airport operator and the holder of the aircraft certificate of registration;
- b) Arrangements for notifying the holder of the certificate of registration;
- c) Arrangements for liaising with the air traffic control unit;
- d) Arrangements for obtaining equipment and personnel to remove disabled aircraft; and
- e) The names, role and telephone numbers of persons responsible for arranging for the removal of disabled aircraft.

When the required equipment is not on the Airport and if there is an agreement with a third party for the supplying of equipment, this agreement shall be stated in the Aerodrome Manual.

The equipment to be used, on the Airport or outside the aerodrome, shall be listed

The Aerodrome Manual shall also specify that:

- a) The site shall be secured;
- b) The aircraft shall not be displaced without the authorization of the appropriate authorities;
- c) Appropriate NOTAM's shall be issued;
- d) The maneuvering area shall be inspected once the aircraft has been removed.

The aircraft owner is responsible for:

- Supplying the technical data;
- Take whatever steps is required to have the aircraft removed; and
- Comply with the Airport operator directives.

4.15. HANDLING OF HAZARDOUS MATERIALS

Particulars of the procedures for the safe handling and storage of hazardous materials on the aerodrome, including the following:

- i) Arrangements for special areas on the Airport to be set up for the storage of inflammable liquids (including aviation fuels) and any other hazardous materials; and
- ii) The method to be followed for the delivery, storage, dispensing and handling of hazardous materials.

Note: - Hazardous materials include inflammable liquids and solids, corrosive liquids, compressed gases and magnetized or radioactive materials. Arrangements for dealing with the accidental spillage of hazardous materials should be included in the Airport emergency plan.

In this section, the Aerodrome Manual shall provide procedures for the safe handling and storage of dangerous goods (excluding aircraft refueling) including:

- iii) Identification of aircraft parking stands for the safe loading/offloading of dangerous goods;
- iv) Identification of aircraft parking stands for long term parking of aircraft

- transporting dangerous goods;
- v) The procedures for informing the Rescue and Fire-Fighting service concerning the dangerous goods being on the Airport or transiting through the aerodrome;
- vi) Description of the installations used for the storage of dangerous goods;
- vii) The identification of Airport areas where dangerous goods are stored;
- viii) The identification of the roads used for transportation of dangerous goods.

4.16. Low VISIBILITY OPERATIONS

Particulars of procedures to be introduced for low-visibility operations, including the measurement and reporting of runway visual range as and when required and the names and telephone numbers during and after working hours of the persons responsible for measuring the runway visual range.

The Aerodrome Manual shall also contain:

- i) Coordination procedures between the Airport operator and the Air Traffic Services;
- ii) The Airport operator procedures to ensure that the Rescue and Fire-Fighting service is on stand-by and that the runway visual range is measured and reported;
- iii) Procedure for ensuring that pilots will not be presented with inadequate visual guidance or misleading information when runway visual ranges are less than 550m.
- iv) The Airport operator procedures to inform, in real time, all staff likely to drive a vehicle on the airside that low visibility procedures are in force; and
- v) Restrictions applicable to the circulation of vehicles on the air side.

N.B: All matters related to electrical power supply shall be dealt with at Item 4.6 «Visual Aids and Airport Electrical Systems».

4.17. PROTECTION OF SITES FOR RADAR, NAVIGATIONAL AIDS AND METEOROLOGICAL EQUIPMENT

Particulars of the procedures for the protection of site for radar and radio navigational aids located on the Airport to ensure that their performance will not

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be degraded including the following:

- i) Arrangements for the control of activities in the vicinity of radar and Navaids installations;
- ii) Arrangements for ground maintenance in the vicinity of these installations;
- iii) process include an assessment of the aerodrome visual aids; and
- iv) Implemented a process to ensure that the siting or performance of visual and non-visual aids to navigation is not adversely affected by objects under the approach surface.
- v) Arrangements for the supply and installation of signs warning of hazardous microwave radiation.
- vi) Measures taken to make sure that all personnel concerned is aware of the precautions to be taken about Navaids;
- vii) Measures taken to ensure that Navaids constraints are taken into account for all Airport development projects and for all works taking place on the aerodrome; and
- viii) Measures taken to ensure coordination between the Airport operator and the air navigation services.

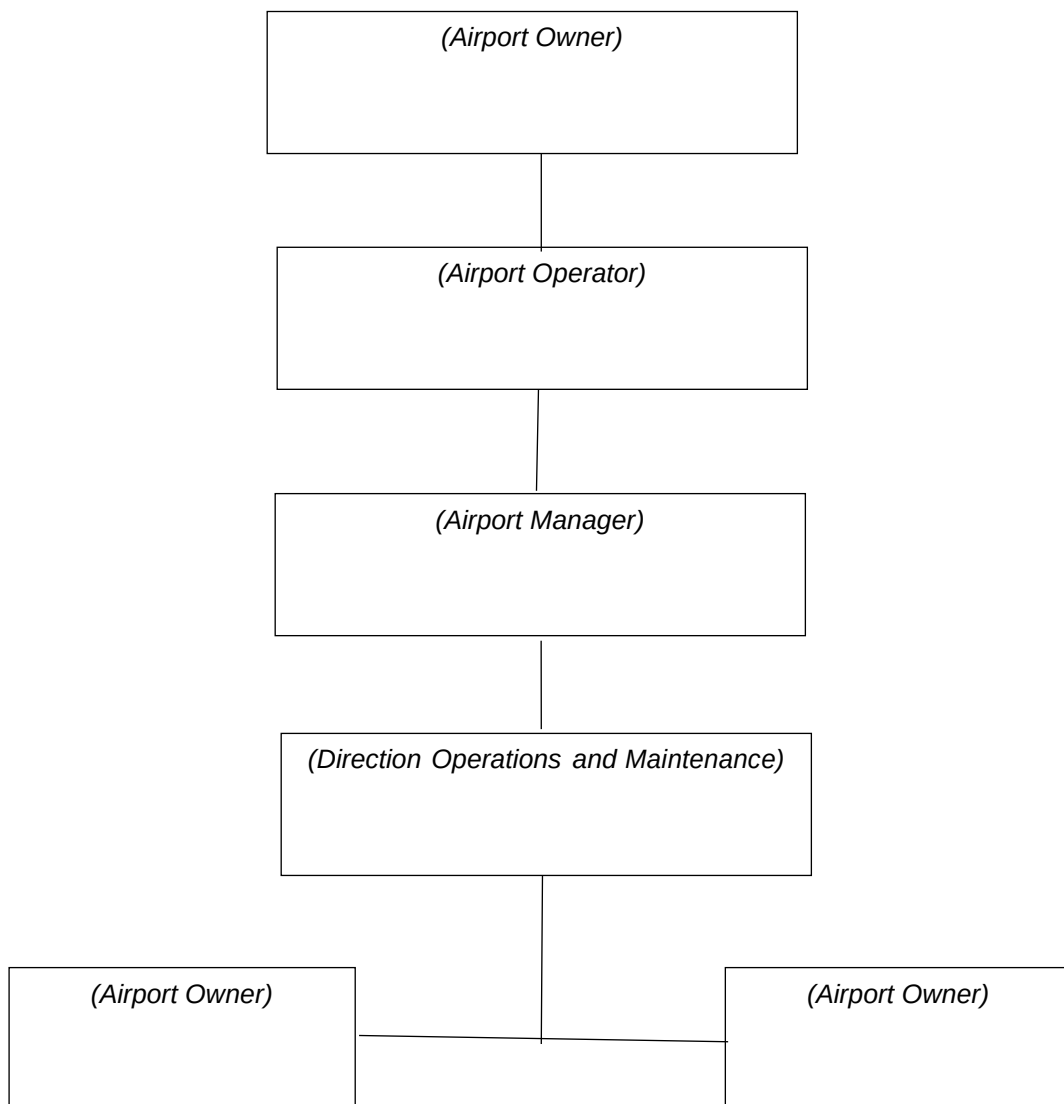
This section shall also include:

- i) A plan identifying the areas to be protected and the associated Navaids;
- ii) The protocols or accords between the Airport operator and the air navigation services.

Chapter 5 - AIRPORT ADMINISTRATION

5.1. INFORMATION ON AIRPORT ADMINISTRATION:

- i) Airport Organizational Chart showing the names, titles and functions of key personnel;



- ii) Name, function and phone number of the person responsible for the Airport general safety.
- iii) Airport committees.

Chapter 6 - SAFETY MANAGEMENT SYSTEM (SMS)

This section shall describe the Airport Safety Management System (SMS). This section shall be based on the Airport operator Safety Management System Manual (SMSM) adapted to the specifics of the aerodrome. Information about the components and the elements of the SMS will also be included in this section. These components and elements describe how the aeronautical safety requirements will be met and how these requirements will improve over time.

The information on the four (4) components and the twelve (12) elements of the SMS as established by the ICAO framework are described below.

6.1. COMPONENT1. SAFETY POLICY AND OBJECTIVES

Element 1. Management commitment and responsibility

The following information shall be included:

- i) Senior management must develop the safety policy of the organization to be signed by the Accountable Executive and including the following commitments:
 - a) Observe all applicable legal requirements and international standards, and best effective practices;
 - b) A clear statement about the provision of the necessary resources for the implementation of the safety policy;
 - c) A safety reporting procedures;
 - d) The safety policy shall clearly indicate which types of operational behaviors are unacceptable;
 - e) To indicate the conditions under which disciplinary action would not apply;
 - f) To communicate the safety policy, with visible endorsement, throughout the organization;
 - g) The development of a formal process to develop a set of safety objectives;
 - h) To publicize and distribute safety objectives.

Element 2. Safety Accountabilities

To supply the following information:

- i) Name and title of the Accountable Executive who has final responsibility in the organization for implementing and maintaining the SMS;
- ii) To confirm that the Accountable Executive has full control of the financial resources required for the operations authorized to be conducted under the Airport certificate;
- iii) To confirm that the Accountable Executive has full control of the human resources required for the operations authorized to be conducted under the Airport certificate;
- iv) To confirm that the Accountable Executive has direct responsibility for the conduct of the organization's affairs;
- v) To confirm that the Accountable Executive has final responsibility over operations authorized to be conducted under the Airport certificate;
- vi) Description of the accountability of all members of management, irrespective of other functions, as well as of employees, with respect to the safety performance of the SMS;
- vii) The description of measures taken to ensure that safety responsibilities, accountabilities and authorities are documented and communicated throughout the organization; and
- viii) The definition of the levels of management with authority to make decisions regarding safety risk tolerability.

Element 3. Appointment of Key Safety Personnel

- i) Name and title of the qualified person appointed to manage and supervise the SMS on a day-to-day basis;
- ii) Job description and responsibilities of the person supervising the SMS on a day-to-day basis; and
- iii) Description of the safety authorities, responsibilities and accountabilities of personnel at all levels of the organization defined and documented.

Element 4. Coordination of Emergency Response Planning

- i) To confirm that an Emergency Response Plan has been developed, proportionate

to the size and complexity of the operations;

- ii) To confirm that the organization coordinate its emergency response/contingency procedures with the emergency/response contingency procedures of other organizations it must interface with during the provision of services; and
- iii) To describe the process to distribute and communicate the coordination procedures to the personnel involved in such interaction.

Element 5. SMS Documentation

One explicit feature of an SMS is that all safety management activities are required to be documented and visible. It follows that documentation is an essential element of a SMS.

SMS documentation must include and make reference to, as appropriate, all relevant and applicable national and international regulations. It must also include SMS-specific records and documentation, such as hazard reporting forms, lines of accountability, responsibility and authority regarding the management of operational safety, and the structure of the safety management organization. It must furthermore document explicit guidelines for records management, including handling, storage, retrieval and preservation. But without doubt, the most important piece of documentation of an SMS is the SMS Manual (SMSM).

The SMSM is a key instrument for communicating the organization's approach to safety to the whole organization. It documents all aspects of the SMS, including the safety policy, objectives, procedures and individual safety accountabilities.

Typical contents of an SMSM include:

- i) Scope of the safety management system;
- ii) Safety policy and objectives;
- iii) Safety accountabilities;
- iv) Key safety personnel;
- v) Documentation control procedures;
- vi) Coordination of emergency response planning;
- vii) Hazard identification and risk management schemes;
- viii) Safety assurance;
- ix) Safety performance monitoring\ safety auditing;

- x) Management of change;
- xi) Safety promotion; and
- xii) Contracted activities.

The SMSM shall meet the following requirements:

- i) To develop and to maintain a safety library for appropriate hazard documentation and documentation management;
- ii) To develop and maintain SMS documentation in paper or electronic form;
- iii) To develop SMS documentation in a manner that describes the SMS and the consolidated interrelationships between all the SMS components;
- iv) To develop an SMS implementation plan that ensures that the SMS meets the organization's safety objectives;
- v) The SMS implementation plan to be developed by a person or a planning group which comprises an appropriate experience base;
- vi) To specify if the person or planning group received enough resources (including time for meetings) for the development of the SMS implementation plan;
- vii) SMS implementation plan to be endorsed by the senior management of the organization;
- viii) SMS implementation plan to be regularly reviewed by the senior management of the organization;
- ix) The SMS implementation plan to implement SMS in phases;
- x) The SMS implementation plan shall explicitly address the coordination between the Airport operator's SMS and the SMS of other organizations the Airport operator must interface with during the provision of services;
- xi) The Airport operator to develop a safety management systems manual (SMSM) as a key instrument for communicating the organization's approach to safety to the whole organization;
- xii) The SMSM to document all aspects of the SMS including, among others, the safety policy, objectives, procedures and individual safety accountabilities;
- xiii) The SMSM to clearly articulate the role of safety risk management as an initial design activity and the role of safety assurance as a continuous activity;
- xiv) The relevant portions of SMS-related documentation shall be incorporated into approved documentation, such as company operations manual, maintenance control/policy manual and Aerodrome Manual, as applicable;

- xv) A records system that ensures the generation and retention of all records necessary to document and support operational requirements;
- xvi) The Airport operator records system to be in accordance with applicable regulatory requirements and industry best practices;
- xvii) The Airport operator records system to provide the control processes necessary to ensure appropriate identification, legibility, storage, protection, archiving, retrieval, retention time, and disposition of records.

6.2. COMPONENT 2. SAFETY RISK MANAGEMENT

Element 1. Hazard Identification

- i) The description of a formal safety data collection and processing for effectively collecting information about hazards in operations;
- ii) The description of the measures to include a combination of reactive, proactive and predictive methods of safety data collection;
- iii) The description of the reactive processes that provide for the capture of information relevant to safety and risk management;
- iv) The description of training relevant to reactive methods of safety data collection;
- v) The description of communication relevant to reactive methods of safety data collection;
- vi) The description of the methods used showing that reactive reporting are simple, accessible and commensurate with the size of the Airport operator's activities;
- vii) The description of the procedure showing that reactive reports are reviewed at the appropriate level of management;
- viii) The description of the feedback process to notify contributors that their reports have been received and to share the results of the analysis;
- ix) The description of the proactive processes that actively look for the identification of safety risks through the analysis of the organization's activities;
- x) The description of the training relevant to proactive methods of safety data collection;
- xi) The description of communication relevant to proactive methods of safety data collection;
- xii) The description of the proactive reporting system showing that it is simple, accessible and commensurate with the size of the Airport operator's business;

- xiii) The description of the predictive processes that provide the capture of system performance as it happens in real-time normal operations;
- xiv) The description of the training relevant to predictive methods of safety data collection;
- xv) The description of the communication developed relevant to predictive methods of safety data collection;
- xvi) The description of the predictive safety data capture process to show that it is commensurate with the size of the Airport operator's business.

Element 2. Safety Risk Assessment and Mitigation

- i) The description of the process to analyze, evaluate and control operational safety risks;
- ii) The description of the formal process to analyze in terms of probability and seriousness the safety risks associated with identified dangers;
- iii) Description of criteria used to evaluate the safety risks and to determine if they are acceptable (i.e. acceptable level of risk the organization can accept);
- iv) Description of safety risks mitigation strategies which include preventive/corrective action plan to prevent the reoccurrence of these risks and deficiencies.

6.3. COMPONENT 3. SAFETY ASSURANCE

Element 1. Safety Performance Monitoring and Measurement

- i) Description of the internal process to verify the safety performance of the organization and to validate the effectiveness of safety risks controls;
- ii) To include the following tools in those processes:
 - a. Safety reporting systems
 - b. Safety studies
 - c. Safety reviews
 - d. Safety audits
 - e. Safety surveys
 - f. Internal safety investigations
- iii) To verify the safety performance of the organization in reference to the safety performance indicators and safety performance targets of the SMS;

- iv) To review safety reports at the appropriate level of management;
- v) Description of the feedback process to notify contributors that their reports have been received and to share the results of the analysis;
- vi) To describe how corrective and preventive actions are generated in response to hazard identification;
- vii) Description of the procedures for the conduct of internal investigations;
- viii) Description of the process to ensure that occurrences and deficiencies reported are analyzed to identify all associated hazards;
- ix) To describe the process for evaluating the effectiveness of the corrective/preventive measures that have been developed;
- x) To describe the system to monitor the internal reporting process and the associated corrective actions;
- xi) Description of the audit function with independence and authority required to carry out effective internal evaluations;
- xii) Description of the audit system covering all functions and activities of the Airport Operator;
- xiii) To describe the selection/training processes to ensure the objectivity and competence of auditors as well as the impartiality of the audit process;
- xiv) To describe the procedure for reporting audit results and maintaining records;
- xv) To describe the procedure for outlining requirements for timely corrective and preventive actions in response to audit results;
- xvi) To describe procedure to record verification of action(s) taken and the reporting of verification results; and
- xvii) To describe the process in place to monitor and analyze trends.

Element 2. The Management of Change

- i) To describe the formal process put in place and maintain by the IACM to identify changes within the organization which may affect established processes and services;
- ii) To describe how the formal process for the management of change analyze changes to operations or key personnel for safety risks;
- iii) To describe the arrangements established to ensure safety performance prior to implementing changes; and
- iv) To describe the process establish to eliminate or modify safety risk controls

that are no longer needed due to changes in the operational environment.

Element 3. Continuous Improvement of the SMS

- i) The description of the process developed and maintain by the IACM to identify the causes of substandard performance of the SMS;
- ii) The description of the mechanism established to determine the implications of substandard performance of the SMS on operations;
- iii) The description of the mechanism established to eliminate or mitigate the causes of substandard performance of the SMS;
- iv) The description of the process for the proactive evaluation of facilities, equipment, documentation and procedures (through audits and surveys, etc.); and
- v) The description of the process for the proactive evaluation of an individual's performance, to verify the fulfilment of that individual's safety responsibilities.

6.4. COMPONENT 4. SAFETY PROMOTION

Element 1. Training and Education

- i) The description of the process to identify training requirements so that personnel are trained and competent to perform their SMS duties;
- ii) The description of the measures put in place to make sure that the training is appropriate to the individual's involvement in the SMS;
- iii) The description of procedures to incorporate safety training into indoctrination training upon employment;
- iv) The description of the emergency response/contingency training for affected personnel; and
- v) The description of the process that measures the effectiveness of training.

Element 2. Safety Communication

- i) The description of the communication processes in place within the organization that permit the SMS to function effectively;
- ii) The description of the communication processes (written, meetings, electronic, etc.) commensurate with the size and scope of the organization;
- iii) To describe how safety-critical information is established and maintained in a

suitable medium that provides direction regarding relevant SMS documents;

- iv) To describe how safety-critical information is disseminated throughout the organization and is the effectiveness of safety communication monitored; and
- v) To describe the procedure that explains why particular safety actions are taken and why safety procedures are introduced or changed.

Chapter 7 - AIRPORT OPERATIONS PROTOCOLS, ACCORDS, CONTRACTS AND CONVENTIONS WITH THIRD PARTIES

The Airport operations protocols, accords, contracts and conventions with third parties shall be included in the Aerodrome Manual.

When Airport operator's tasks are entrusted, in whole or in part, to a third party, the Airport operator shall describe the conditions he put in place to ensure that the third party will conform to existing laws and regulations and will adopt adequate operational procedures.

The Airport operator should identify which service of the third party is responsible for any specific task.

Chapter 8 - APPENDICES

8.1. PLANS

To include all plans mentioned in the Aerodrome Manual.

8.2. AIRPORT EMERGENCY PLAN

This manual is not included in the present document because of its volume and of its operational nature. It is available from the Airport manager, if required and appropriate.

Most of the Aerodrome Manual holders already have a copy of the Airport Emergency Plan.

8.3. OTHER APPENDICES

To include all other appendices mentioned in the Aerodrome Manual.